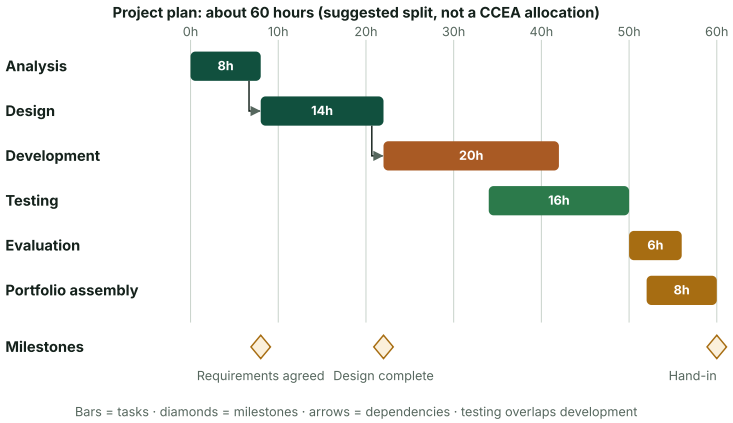
Understanding the task: case study, planning and authentication

AD01 ANSWERS

I can: Describe what the A2 2 task involves and how the portfolio is organised · Produce a realistic Gantt chart for a 60-hour project · Explain the rules for authentication and how to acknowledge help and sources



Use this diagram to help you: A Gantt chart for a 60-hour database project showing six tasks, overlapping testing, dependencies and three milestones

A Key terms

Write the letter of the matching definition · 6 marks

F Case study

B Gantt chart

E Milestone

C Dependency

D Critical path

A Authentication

A Confirmation by the candidate and teacher that coursework is the candidate's own work, produced under the required conditions.

B A planning chart showing each task as a horizontal bar against a timeline, with durations, overlaps and dependencies.

C A relationship between tasks in which one task cannot start until another has been completed.

D The longest sequence of dependent tasks in a project; a delay to any task on it delays the whole project.

E A significant checkpoint in a project, such as the completion of a stage, used to monitor progress.

F A scenario pre-released by CCEA describing an organisation's problem, which candidates analyse and solve with a database application.

B Fill the gaps

Use the word bank · 6 marks

case study | teacher | database | Gantt chart | milestone | authenticated

A2 2 is assessed by your **teacher** and moderated by CCEA. You build a relational **database** application in response to a pre-released **case study** . A **Gantt chart** shows tasks against time, and a **milestone** marks an important checkpoint. All help and sources must be acknowledged so the work can be **authenticated**.

C True or false?

Tick one box · 6 marks

	True	False
A2 2 evidence is submitted as a single PDF portfolio.	☑	☐
The case study is the same every year, so last year's notes can be reused. *(CCEA releases a case study for the series; always use the current one and its instructions.)*	☐	☑
A dependency means one task cannot start until another has finished.	☑	☐
You do not need to acknowledge online tutorials if you changed the wording. *(All sources, tutorials, templates and help must be acknowledged.)*	☐	☑
Updating your Gantt chart after a delay is good evidence of project management.	☑	☐
The moderator can view your database file, so screenshots are optional. *(Credit is based on the portfolio evidence, so annotated screenshots are essential.)*	☐	☑

D Exam-style questions

20 marks

1 State what is produced in the A2 2 Application Development task and how the evidence is submitted. [2]

> A relational database application that solves the problem in CCEA's pre-released case study (1) / all evidence (analysis, design, development, testing, evaluation) is submitted as a single PDF portfolio (1).

2 Explain why a Gantt chart is useful when planning a 60-hour project. [3]

> A Gantt chart shows each task as a bar against a timeline, so durations and overlaps can be seen (1) / it shows dependencies, i.e. which tasks cannot start until another finishes, e.g. forms need tables first (1) / and milestones, so progress can be checked against the plan and slippage spotted and managed within the 60 hours / hand-in date (1).

3 Define the term milestone and give an example from a database project. [2]

> A milestone is a key point or checkpoint in a project that marks the completion of a significant stage (it has no duration) (1) / example: requirements agreed with the client / design signed off / test plan complete / database tables and relationships built (1).

4 Describe two things a candidate must do to ensure their coursework can be authenticated. [4]

> Any two, each stated (1) and expanded (1): carry out most of the work under supervision in school (1) / so the teacher can confirm it was produced by the candidate under the required conditions (1) / sign the candidate declaration (1) / confirming the work is their own; the teacher also signs (1) / acknowledge every source, tutorial, template, help and any permitted AI use (1) / in a references/acknowledgements section, so nothing is passed off as their own (1) / keep dated versions of the database and portfolio and a project log (1) / to show how the work developed over time (1).

5 Explain why requirement numbers should be used throughout the portfolio. [3]

> Requirement numbers link each requirement from analysis through design, development, testing and evaluation (1) / so the moderator can trace any requirement and quickly find the evidence that it has been designed, built and tested (1) / and ensure every requirement is tested and evaluated as fully, partially or not met, with nothing missed (e.g. using a traceability table) (1).

6 A candidate plans to spend 40 of the 60 hours on development and leave testing and evaluation until the last week. Evaluate this plan. [6]

> Indicative content. Problems: 40 hours on development is double the suggested 20 hours, leaving only 20 hours for analysis, design, testing, evaluation and the portfolio / suggests too little time on analysis and design – weak requirements and design make development slower and testing/evaluation harder / leaving testing to the last week leaves no time to fix failed tests and retest, or to carry out user acceptance testing / evaluation will be

rushed and may not cover every requirement / any slippage means no contingency before the hand-in date / development evidence should be captured as you go; testing should happen iteratively as each part is built. Possible strengths: development is the largest stage and building a complex solution takes time; some testing will happen informally during development. Improved plan: roughly analysis 8 h, design 14 h, development 20 h, testing 10 h, evaluation and portfolio 8 h; write the test plan during design; test each component as it is built; include milestones and contingency; keep a project log. Conclusion: the plan is poorly balanced and risky. Level 3 (5–6): balanced discussion of both benefits and drawbacks, well applied to the scenario, with a reasoned conclusion; QWC: well organised, accurate specialist terms. Level 2 (3–4): several relevant points, some application, may be one-sided or lack a conclusion. Level 1 (1–2): a few simple points, little or no application or explanation.

E Quick quiz

Circle one answer · 5 marks

1 What does the A2 2 task require you to produce? [1]

A A game written in C#
B A relational database application ✓
C A website
D A written essay

2 Which shows tasks as bars against a timeline? [1]

A Data flow diagram
B ER diagram
C Gantt chart ✓
D Wireframe

3 What is a milestone? [1]

A A key checkpoint in the project ✓
B A type of query
C A field in a table
D An error message

4 Which of these is malpractice? [1]

A Citing a tutorial you used
B Signing the authentication declaration
C Keeping a project log
D Submitting another pupil's design as your own ✓

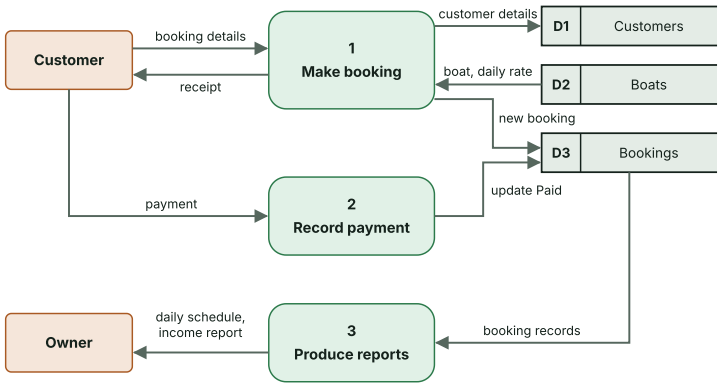
5 Where should you check the marking criteria? [1]

A A revision guide from 2017
B The current CCEA marking grid ✓
C A friend's portfolio
D The software help menu

Analysis: client, current system and requirements

AD02 ANSWERS

I can: Plan and record a client interview that gathers useful detail · Write numbered, measurable functional and non-functional requirements · Draw context and level 1 data flow diagrams of the proposed system



Rectangle = external entity · rounded box = process · open box = data store · arrow = data flow

Use this diagram to help you: A level 1 data flow diagram for the proposed Erne Boat Hire system with two external entities, three processes and three data stores

A Key terms

Write the letter of the matching definition · 6 marks

F Requirements specification

D Functional requirement

E Non-functional requirement

B Measurable requirement

C Data flow diagram

A Context diagram

A A top-level data flow diagram showing the whole system as one process together with its external entities and data flows.

B A requirement with a specific, testable target so that it can be judged objectively as met or not met.

C A diagram that models how data moves between external entities, processes and data stores in a system.

D A statement of what the system must do, such as the data it stores, the processes it carries out or the outputs it produces.

E A statement of how well the system must perform, covering qualities such as usability, performance, security or reliability.

F A numbered, agreed list of the functional and non-functional requirements that the new system must meet.

B Fill the gaps

Use the word bank · 5 marks

external entity | functional | non-functional | measurable | data store

A **functional** requirement states what the system must do, while a **non-functional** requirement states how well it must do it. Every requirement should be numbered and **measurable**. In a DFD, a person or organisation outside the system is an **external entity**, and data held by the system is shown as a **data store**.

C True or false?

Tick one box · 6 marks

	True	False
'Search results must appear in under 3 seconds' is a non-functional requirement.	☑	☐
'The system should be fast' is a good requirement because everyone understands it. *(It is not measurable; give a time and a condition.)*	☐	☑
Open questions are useful for exploring how the current system works.	☑	☐
In a DFD, data can flow directly from one data store to another. *(Data must pass through a process between two data stores.)*	☐	☑
Every problem with the current system should lead to at least one requirement.	☑	☐
A context diagram shows every data store in the system. *(A context diagram shows the system as a single process with external entities; data stores appear at level 1.)*	☐	☑

D Exam-style questions

19 marks

1 Distinguish between a functional and a non-functional requirement, giving an example of each for a boat hire booking system. [4]

> A functional requirement states what the system must do – the data, processes and outputs (1) / e.g. the system must calculate the total cost as number of days × daily rate of the boat (1) / A non-functional requirement states how well the system must perform – usability, performance, security, reliability (1) / e.g. only staff with a valid username and password can open the booking forms, or any form reachable from the main menu in no more than two clicks (1).

2 Explain why the requirement 'The system should be user-friendly' is weak and rewrite it. [3]

> It is vague / subjective – 'user-friendly' means different things to different people (1) / so it cannot be measured or tested objectively, and cannot be evaluated as met or not met (1) / rewritten, e.g.: 'A new member of staff can add a booking in under 2 minutes after a 10-minute demonstration' or 'Every form can be reached from the main menu in no more than two clicks' (1).

3 Give one advantage and one disadvantage of interviewing the client as a fact-finding method. [2]

> Advantage, any one: allows follow-up questions to clarify answers / gives detailed, in-depth information about the client's needs (1) / Disadvantage, any one: time-consuming to arrange, carry out and write up / gives only one person's view, which may be biased or incomplete (1).

4 State the purpose of each DFD symbol: external entity, process, data store, data flow. [4]

> External entity: a person or organisation outside the system that is a source or destination of data, e.g. Customer (1) / Process: transforms or acts on data, e.g. 1 Make booking (1) / Data store: holds data at rest within the system, e.g. D1 Customers (1) / Data flow: shows data moving between entities, processes and stores, labelled with the data, e.g. 'booking details' (1).

5 Describe the difference between a context diagram and a level 1 DFD. [2]

> A context diagram shows the whole system as a single process with the external entities and the data flows between them (1) / a level 1 DFD breaks that process down into the main sub-processes and shows the data stores and the flows between them (1).

6 Explain how a well-written requirements specification helps in the testing and evaluation stages of the project. [4]

> Any four from: each requirement is specific, numbered and measurable (1) / so tests can be written directly from it, with precise test data and expected results (1) / ensures every requirement is covered by at least one test; nothing is missed (1) / the requirement numbers allow the test plan to be cross-referenced/traced (1) / in the evaluation, each requirement can be judged as fully, partially or not met (1) / supported by objective evidence

such as test numbers and user feedback rather than opinion (1) / and partially/not met requirements form the basis of justified improvements (1).

E Quick quiz

Circle one answer · 5 marks

1 Which is a functional requirement? [1]

A Results appear in 3 seconds

B The system must calculate the total cost of a booking ✓

C Only staff can log in

D The menu uses a large font

2 Which fact-finding method allows follow-up questions? [1]

A Questionnaire

B Document analysis

C Interview ✓

D Observation

3 What makes a requirement measurable? [1]

A It uses the word 'should'

B It has a clear target that can be tested ✓

C It is written by the client

D It mentions the software

4 In a DFD, what does an open-ended box labelled D1 show? [1]

A Data store ✓

B External entity

C Process

D Data flow

5 Which diagram shows the whole system as a single process? [1]

A Level 1 DFD

B ER diagram

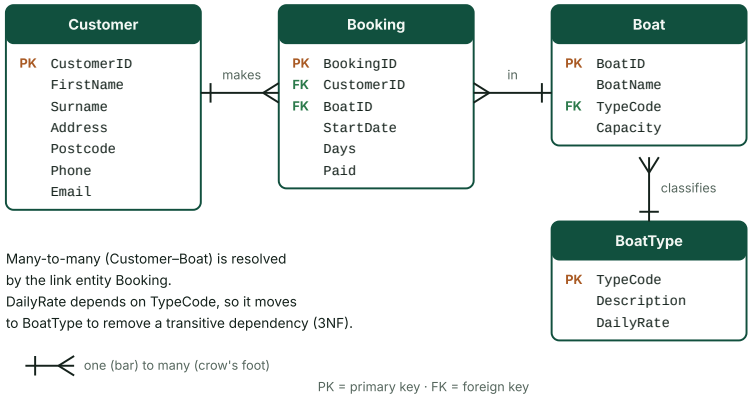
C Gantt chart

D Context diagram ✓

Design: ER diagrams, normalisation, data dictionary and interfaces

AD03 ANSWERS

I can: Draw an ER diagram and normalise data to third normal form · Produce a complete data dictionary with validation and keys · Design wireframes, queries and pseudocode algorithms linked to requirements



Use this diagram to help you: An ER diagram for Erne Boat Hire in third normal form: Customer and Boat each have a one-to-many relationship with Booking, and BoatType has a one-to-many relationship with Boat

A Key terms

Write the letter of the matching definition · 6 marks

F Entity relationship diagram

C Foreign key

B Normalisation

E Third normal form

D Data dictionary

A Wireframe

A A labelled sketch of the layout of a form, report or screen showing its controls and navigation before it is built.

B The process of organising data into related tables to remove redundancy and prevent update, insertion and deletion anomalies.

C A field in one table that is the primary key of another table, used to create a relationship between them.

D A document recording each field's name, data type, size, validation rule, key status and description.

E A table is in 3NF when it is in 2NF and no non-key attribute depends on another non-key attribute.

F A diagram showing the entities in a system and the one-to-one, one-to-many or many-to-many relationships between them.

B Fill the gaps

Use the word bank · 5 marks

third normal form | foreign key | ER diagram | link entity | data dictionary

An **ER diagram** shows entities and the relationships between them. A many-to-many relationship is resolved using a **link entity**. Data in **third normal form** has no partial or transitive dependencies. A **foreign key** links a table to the primary key of another table. The **data dictionary** records the type, size and validation of every field.

C True or false?

Tick one box · 6 marks

	True	False
A primary key uniquely identifies each record in a table.	☑	☐
Relational databases implement many-to-many relationships directly without a link table. *(A link table is used to create two one-to-many relationships.)*	☐	☑
A transitive dependency is removed when moving from 2NF to 3NF.	☑	☐
'Range check' is enough detail for a validation rule in a data dictionary. *(Give the actual rule, such as between 1 and 14.)*	☐	☑
Wireframes should be annotated with controls, validation and navigation.	☑	☐
Normalisation increases data redundancy. *(Normalisation reduces redundancy.)*	☐	☑

D Exam-style questions

22 marks

1 Define the term foreign key. [2]

> A foreign key is a field (attribute) in one table that is the primary key of another table (1) / used to create the relationship between the two tables and allow referential integrity to be enforced, e.g. CustomerID in Booking (1).

2 Explain why a many-to-many relationship between Customer and Boat must be resolved, and how it is resolved. [3]

> A customer can hire many boats and a boat can be hired by many customers over time (1) / a relational database cannot implement M:N directly because a field cannot hold a list of keys – fields must be atomic (1) / it is resolved by a link entity, Booking, which holds CustomerID and BoatID as foreign keys, giving two one-to-many relationships: Customer 1–M Booking and Boat 1–M Booking (1).

3 A Boat table contains BoatID, BoatName, TypeCode and DailyRate, where the daily rate depends on the type of boat. Explain which normal form is not satisfied and how to correct it. [4]

> Third normal form is not satisfied (1) / because there is a transitive dependency: DailyRate depends on TypeCode, which is a non-key attribute, rather than directly on the key BoatID (1) / correct it by moving DailyRate into a new table BoatType(TypeCode, Description, DailyRate) (1) / and leaving Boat(BoatID, BoatName, TypeCode*) with TypeCode as a foreign key linking to BoatType (1).

4 State four items that should be recorded in a data dictionary for each field. [4]

> Any four from: field name (1) / data type (1) / field size / length / format (1) / validation rule (1) / primary or foreign key (key status) (1) / description / example data (1) / default value / required (presence) (1) / input mask (1).

5 Describe what should be included in an annotated wireframe for a data entry form. [3]

> Any three from: layout of all controls (labels, text boxes, combo boxes, buttons) with labels (1) / validation rules and error messages for each field (1) / default values and drop-down lists / combo box contents (1) / navigation buttons, e.g. back to main menu, save, close (1) / the data source (table or query) for the form (1) / the requirement numbers it meets (1) / consistent font, colour scheme and accessibility suited to the users (1).

6 Discuss the importance of the design stage in producing a high-quality database application. [6]

> Indicative content. Design is a blueprint detailed enough for someone else to build the system / a correct ER diagram resolves M:N relationships and identifies keys, so tables link correctly / normalising to 3NF removes redundancy and anomalies (e.g. DailyRate in BoatType), keeping data consistent / the data dictionary specifies types, sizes and precise validation, so data quality is built in and development is faster with fewer fixes / annotated wireframes ensure forms and reports suit the users' IT skills and can be checked with the client before building / query designs and pseudocode for calculations, macros or VBA reduce errors in development / linking every design item to requirement numbers ensures all requirements are met and makes testing and evaluation straightforward; a test strategy can be planned / design errors are much cheaper to fix than errors

found after building. Counterpoints: too long on design leaves less time for development; designs often change during development, but changes should be noted and justified. Conclusion: good design is the foundation of a high-quality, maintainable application. Level 3 (5–6): balanced discussion of both benefits and drawbacks, well applied to the scenario, with a reasoned conclusion; QWC: well organised, accurate specialist terms. Level 2 (3–4): several relevant points, some application, may be one-sided or lack a conclusion. Level 1 (1–2): a few simple points, little or no application or explanation.

E Quick quiz

Circle one answer · 5 marks

1 Which normal form removes repeating groups? [1]

A First normal form ✓ B Second normal form
C Third normal form D Unnormalised form

2 How is a many-to-many relationship resolved? [1]

A Delete one entity **B Add a link entity ✓**
C Use a larger primary key D Merge the tables

3 Which is a transitive dependency? [1]

A BookingID → Days B CustomerID → Surname
C TypeCode → DailyRate in the Boat table ✓ D BoatID → BoatName

4 Which item does NOT belong in a data dictionary? [1]

A Data type B Field size
C Validation rule **D The client's opinion of the colour scheme ✓**

5 What should be designed in pseudocode? [1]

A The ER diagram **B Any code or complex macro ✓**
C The Gantt chart D The title page

Development: building and evidencing the database application

AD04 ANSWERS

I can: Build tables and relationships that match the design and enforce referential integrity · Create validated forms, parameter and calculated queries, reports and a menu · Produce annotated screenshot evidence that links each feature to the design and requirements

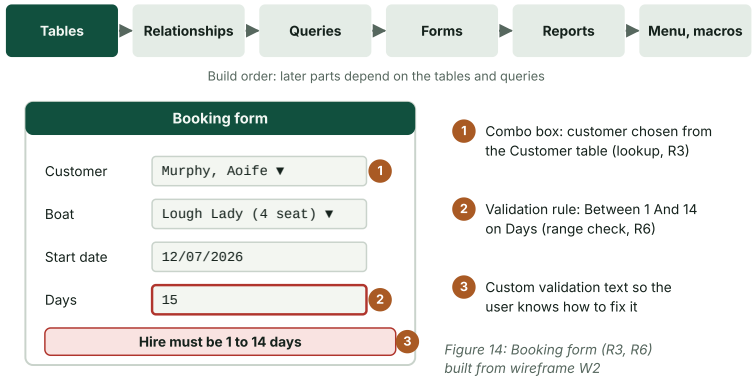


Figure 14: Booking form (R3, R6) built from wireframe W2

Crop, call out, caption, link to requirement

Use this diagram to help you: The order for building a database application, and an annotated screenshot of a booking form showing a lookup, a range check and its error message

A Key terms

`Write the letter of the matching definition · 6 marks`

F Referential integrity

D Parameter query

B Calculated field

E Input mask

C Macro

A Annotated screenshot

A A cropped screen capture with labels and a caption explaining what it shows and which requirement and design item it evidences.

B A field in a query, form or report whose value is derived from other fields, such as cost from days and daily rate.

C A stored sequence of actions that automates a task in a database, such as opening a form or printing a report.

D A query that prompts the user to enter a value when it runs and uses that value as its criteria.

E A pattern that controls the format of data entered into a field, such as a postcode or phone number.

F A rule ensuring every foreign key value matches an existing primary key value in the related table, preventing orphan records.

B Fill the gaps

`Use the word bank · 5 marks`

referential integrity | foreign | parameter query | calculated field | annotated

Tables are linked using primary and **foreign** keys, and **referential integrity** prevents orphan records. A **parameter query** asks the user for a value when it runs. A **calculated field** derives a value such as cost from other fields. Every feature should be evidenced with an **annotated** screenshot linked to a requirement.

C True or false?

Tick one box · 5 marks

	True	False
Referential integrity stops a booking being saved for a customer who does not exist.	☑	☐
A calculated field should be stored as a separate field in the table. *(Calculated values are derived when needed, not stored, to avoid redundancy.)*	☐	☑
A parameter query can be reused for different dates without editing it.	☑	☐
Validation guarantees that the data entered is correct. *(Validation checks data is sensible or reasonable, not that it is correct.)*	☐	☑
Showing a problem you found and fixed is useful development evidence.	☑	☐

D Exam-style questions

19 marks

1 Explain what is meant by referential integrity and why it is important in a booking database. [3]

> Referential integrity means every foreign key value must match an existing primary key value in the related table (1) / e.g. a booking cannot be entered for a CustomerID or BoatID that does not exist (no orphan records) (1) / important because bookings must always link to a real customer and boat, so reports, receipts and costs are accurate and the data is consistent; it also prevents a customer with bookings being deleted (if restrict is set) (1).

2 Describe what a parameter query is and give an example from a boat hire system. [3]

> A parameter query prompts the user to enter a value (criterion) when it is run (1) / so one query can be reused for any value instead of creating a separate query for each (1) / example: a query on Booking and Customer with criteria StartDate = [Enter date:] to list all bookings on a date; or search by customer surname (1).

3 Explain why the total cost of a booking should be produced by a calculated field rather than stored in the Booking table. [2]

> The total cost is derived data – it can be calculated from Days × DailyRate whenever needed (1) / storing it would create redundancy and risk inconsistency, e.g. if Days or the daily rate is changed the stored total would be wrong (calculated fields always give the current correct value) (1).

4 Give three validation checks that could be applied to fields in a Booking form, stating the field and the rule. [3]

> Any three, each with field and rule (1 each): Days – range check Between 1 And 14 (1) / StartDate – presence check and >= Date() / type check (date) (1) / CustomerID – lookup: chosen from a combo box listing existing customers (1) / BoatID – lookup from a combo box of boats (1) / Paid – Boolean Yes/No (1) / any required field – presence check, e.g. Required = Yes (1).

5 Describe two features of a good annotated screenshot in a development section. [4]

> Any two features, each stated (1) and expanded (1): cropped and readable (1) / showing only the relevant part, so the text is legible at normal zoom (1) / callouts/arrows highlighting the key settings (1) / e.g. pointing to the validation rule or join, with a brief explanation of what it does and why (1) / a figure number and caption (1) / naming the requirement and design item it implements, e.g. 'Figure 14: Booking form (R3, R6) built from wireframe W2' (1) / shows both setup and outcome (1) / e.g. design view with property sheet and the form in use (1) / shows iterative development (1) / a problem found, the change made and why it is better (1).

6 A candidate enforces referential integrity with cascade delete between Customer and Booking. Discuss whether this is appropriate. [4]

> Points for and against (any four): cascade delete means deleting a customer automatically deletes all their bookings (1) / this keeps referential integrity and avoids orphan bookings, and makes it easy to remove a customer completely, e.g. on a valid erasure request (1) / but it risks losing the booking history needed for financial records, income reports and auditing (1) / an accidental deletion would silently remove many records (1) / so restrict delete (refuse to delete customers with bookings) is usually more appropriate, perhaps with an

'archived/inactive' flag on the customer instead (1) / cascade update may still be appropriate if CustomerID can change (1). Conclusion: not usually appropriate unless justified for the client (1).

E Quick quiz

Circle one answer · 5 marks

1 What does enforcing referential integrity prevent? [1]

A Duplicate field names
B Orphan records ✓
C Slow queries
D Large file sizes

2 Which criteria creates a parameter query? [1]

A > 10
B Between 1 And 14
C [Enter surname:] ✓
D Is Null

3 Which is a calculated field? [1]

A Cost: [Days]*[DailyRate] ✓
B Surname
C BookingID
D Postcode

4 What is the best order to build in? [1]

A Reports, forms, tables
B Menu, queries, tables
C Forms, tables, relationships
D Tables, relationships, queries, forms, reports ✓

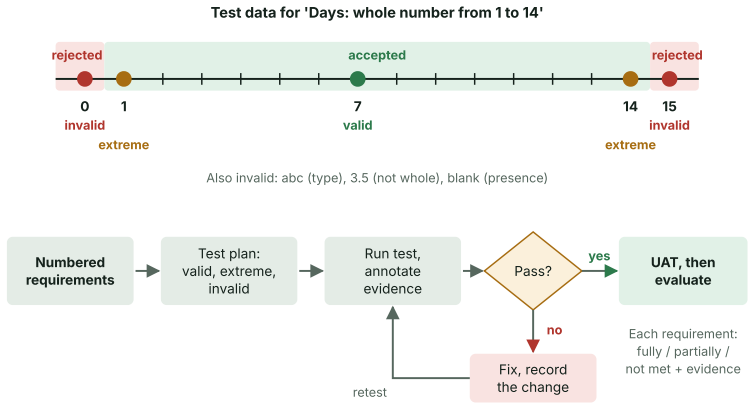
5 Which caption is the strongest evidence? [1]

A Screenshot 4
B My form
C Figure 14: Booking form (R3, R6) from wireframe W2 ✓
D Done

Testing and evaluation against the requirements

AD05 ANSWERS

I can: Write a test plan that covers every requirement with valid, invalid and extreme data · Present annotated test evidence including fixes, retests and user acceptance testing · Evaluate each requirement as fully, partially or not met and suggest justified improvements



Use this diagram to help you: Choosing valid, extreme and invalid test data for a 1 to 14 range, and the cycle from requirements through testing, fixing and retesting to user acceptance testing and evaluation

A Key terms

Write the letter of the matching definition · 6 marks

F **Test plan**	A Data that is outside the acceptable range or of the wrong type, which the system should reject.
C **Valid data**	B Data at the boundaries of the acceptable range, which the system should accept; also called boundary data.
B **Extreme data**	C Normal data within the acceptable range that the system should accept.
A **Invalid data**	D A judgement of how well the finished system meets each requirement, supported by evidence, with justified improvements.
E **User acceptance testing**	E Testing by the client or end users with realistic tasks to confirm the system meets their requirements.
D **Evaluation**	F A planned list of tests, each with a number, requirement, purpose, test data, data type and expected result.

B Fill the gaps

Use the word bank · 5 marks

test plan | user acceptance | extreme | invalid | met

A **test plan** lists each test with its data and expected result. Data at the edge of the acceptable range is **extreme** or boundary data, and data that should be rejected is **invalid** data. Testing by the client is called **user acceptance** testing. In the evaluation each requirement is judged fully, partially or not **met** .

C True or false?

Tick one box · 6 marks

	True	False
For a range of 1 to 14, the value 14 is extreme data and should be accepted.	☑	☐
Only validation needs to be tested in a database project. *(Calculations, queries, reports, navigation and non-functional requirements also need testing.)*	☐	☑
A failed test that is fixed and retested is useful evidence.	☑	☐
'The system is good' is an evaluative statement. *(It has no requirement number, judgement or evidence.)*	☐	☑
User acceptance testing is carried out by the client or end users.	☑	☐
Expected results are written after the tests are carried out. *(Expected results are planned before testing; actual results are recorded during testing.)*	☐	☑

D Exam-style questions

22 marks

1 A field must hold a whole number of days from 1 to 14. Give one example each of valid, extreme and invalid test data. [3]

> Valid (normal): any whole number from 2 to 13, e.g. 5 or 10 (1) / Extreme (boundary): 1 or 14 (1) / Invalid: e.g. 0, 15, -3, 2.5 or 'abc' (1).

2 State four columns that should appear in a test plan. [4]

> Any four from: test number (1) / requirement number (1) / purpose / description of test (1) / test data (1) / type of test data (valid, extreme, invalid) (1) / expected result (1) / actual result (1) / evidence reference (e.g. figure number) (1) / action taken / pass or fail (1).

3 Explain why a test plan should be mapped to the numbered requirements. [2]

> So that every requirement is shown to be tested by at least one test and none is missed (1) / and the test results can be used as evidence in the evaluation to judge each requirement as fully, partially or not met (1).

4 Describe what is meant by user acceptance testing and why it is carried out. [3]

> User acceptance testing is when the client or a representative end user carries out realistic tasks with the finished system, e.g. taking a phone booking (1) / and gives feedback, e.g. on a feedback form linked to the requirements (1) / it is carried out to confirm the system meets the client's needs in real use, especially usability, giving independent evidence for the evaluation and identifying improvements (1).

5 A candidate writes: 'The search works well and the client liked it.' Explain why this is a weak evaluation and rewrite it. [4]

> Why weak, any two: it is vague and subjective – 'works well' is not measured (1) / it gives no evidence, e.g. test numbers, figures or recorded feedback (1) / it is not linked to a numbered requirement or judged fully / partially / not met (1). Rewrite, e.g.: 'R9 (search by surname in under 3 seconds) is fully met (1) / tests 12–14 show that searching for "Maguire" with 5000 bookings returned 3 correct records in under 1 second (Figure 28), and the client's feedback form rated the search 5/5 (1).' Award up to 2 for the rewrite: references a requirement with a judgement (1) / supported by specific evidence (1).

6 Evaluate the importance of testing with extreme and invalid data, as well as valid data, in a database application. [6]

> Indicative content. Valid data shows the system works for typical use, but alone it does not show the system rejects bad data. Extreme/boundary data (e.g. 1 and 14 days) checks the limits of validation rules are set correctly; off-by-one errors (e.g. < 14 instead of <= 14) are common and only boundary tests reveal them. Invalid data (0, 15, wrong type, blank) checks that bad data is rejected with a helpful message, protecting data integrity – invalid data in tables leads to wrong costs, reports and decisions. Testing just outside the boundaries

confirms the rule is exact. Testing all types gives evidence for the evaluation and moderator that validation requirements are fully met, and builds client confidence. Limitations: validation testing cannot prove data is correct, only reasonable; testing should also cover calculations, queries, reports, navigation and non-functional requirements, not just validation. Conclusion: all three types are essential for a robust, reliable database application. Level 3 (5–6): balanced discussion of both benefits and drawbacks, well applied to the scenario, with a reasoned conclusion; QWC: well organised, accurate specialist terms. Level 2 (3–4): several relevant points, some application, may be one-sided or lack a conclusion. Level 1 (1–2): a few simple points, little or no application or explanation.

E Quick quiz

Circle one answer · 5 marks

1 For 'Days from 1 to 14', which value is extreme data? [1]

A 7 — **B 14 ✓**

C 15 — D abc

2 Which is invalid data for 'Days from 1 to 14'? [1]

A 0 ✓ — B 1

C 7 — D 14

3 Who carries out user acceptance testing? [1]

A The moderator — B The developer only

C The client or end users ✓ — D CCEA

4 What should support each evaluation judgement? [1]

A Personal opinion — B The length of the section

C The software used — **D Evidence such as test numbers and feedback ✓**

5 A requirement that works for most cases but not all is... [1]

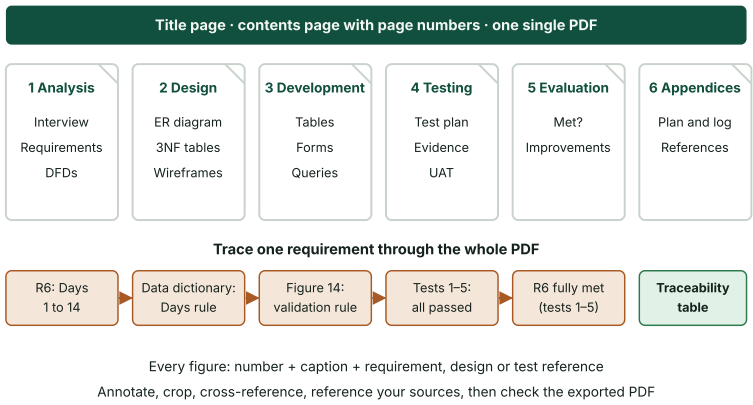
A Partially met ✓ — B Fully met

C Not met — D Invalid

Producing the portfolio

AD06 ANSWERS

I can: Structure a PDF portfolio with a contents page, numbered sections and figures · Annotate and cross-reference screenshots so evidence is easy to find · Reference sources correctly and check the portfolio against a final checklist



Use this diagram to help you: The structure of the single PDF portfolio in six sections, and how one requirement is traced through every section by cross-references

A Key terms

Write the letter of the matching definition · 4 marks

D Portfolio

B Cross-reference

C Traceability table

A Referencing

A Acknowledging the sources used in a piece of work in a consistent format, so it is clear which work is the candidate's own.

B A reference from one part of the portfolio to another, such as a test number or figure number, so evidence can be traced.

C A table linking each requirement to its design, development, test and evaluation evidence.

D The single PDF document containing all the candidate's evidence of analysis, design, development, testing and evaluation.

B Fill the gaps

Use the word bank · 5 marks

caption | contents | cross-reference | PDF | references

The portfolio is submitted as a single **PDF** . A **contents** page with page numbers helps the moderator navigate. Every screenshot needs a figure number and a **caption** . Requirement and test numbers are used to **cross-reference** evidence. All sources are listed in the **references** section.

C True or false?

Tick one box · 5 marks

	True	False
The portfolio should include a contents page with page numbers.	☑	☐
Copying textbook theory into the portfolio adds to your evidence. *(Evidence should be about your own system; copied theory adds nothing and may be plagiarism.)*	☐	☑

Screenshots should be cropped so that their text is readable.	☑	☐
You only need to reference sources you quoted word for word. *(All sources, tutorials, templates and permitted AI use must be acknowledged.)*	☐	☑
You should check the PDF after exporting it.	☑	☐

D Exam-style questions

14 marks

1 State three features that make a PDF portfolio easy for a moderator to navigate. [3]

> Any three from: a title page (1) / a contents page with accurate page numbers (1) / numbered sections following the stages, e.g. 1 Analysis to 5 Evaluation (1) / PDF bookmarks or a linked contents page (1) / consistent headings, figure numbers and captions (1) / clearly labelled appendices (1) / page numbers on every page (1).

2 Describe how a screenshot should be presented as evidence of development. [3]

> Any three from: cropped to the relevant part and readable at normal zoom (1) / annotated with callouts or arrows pointing to the key setting, e.g. a validation rule (1) / a figure number and caption naming the requirement, design item or test number (1) / a sentence or two explaining what it shows and why it was done (1) / show both design view/setup and the result (1).

3 What is a traceability table and why is it useful? [2]

> A traceability table lists each requirement with references to where it is designed, developed and tested (and evaluated) (1) / it is a final check that no requirement has been missed and lets the moderator quickly trace and find the evidence for any requirement (1).

4 Explain why all sources, including tutorials and templates, must be referenced in the portfolio. [2]

> To acknowledge other people's work, so the candidate does not pass it off as their own – otherwise it is plagiarism / malpractice (1) / it is part of authenticating the work: the teacher and moderator can see what is the candidate's own and credit it fairly; CCEA rules require it (1).

5 Give two common presentation mistakes in coursework portfolios and how to avoid them. [4]

> Two mistakes (1 each) with how to avoid (1 each), e.g.: unreadable, uncropped full-screen screenshots (1) / crop, zoom and annotate them with callouts and captions (1) / no contents page or wrong page numbers after final export (1) / check the contents page after exporting the final PDF (1) / screenshots with no explanation or link to requirements (1) / add figure numbers and captions referencing requirement/test numbers (1) / generic theory copied from websites (padding) (1) / remove it and write only about the client's system in your own words (1) / pages out of order or wrong file submitted (1) / open and check the final PDF before submission (1) / spelling and technical errors (1) / proofread (1).

E Quick quiz

Circle one answer · 5 marks

1 How is the A2 2 portfolio submitted? [1]

A As a single PDF ✓ B As the database file only
C As a printed folder only D As a video

2 What should every screenshot have? [1]

A A border **B A figure number and caption ✓**
C A background colour D A title page

3 What does a traceability table show? [1]

A The project budget B The order of slides
C Each requirement's design, development, test and evaluation references ✓ D A list of software

4 Which must be acknowledged? [1]

A Only books B Only quotations

C Nothing

D Tutorials, websites, templates and permitted AI use ✓

5 When should the contents page be checked? [1]

A Before starting analysis

B After the final export ✓

C Only if the moderator asks

D Never